

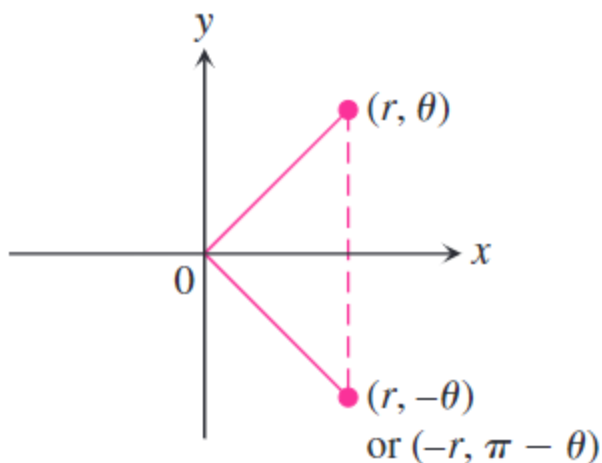
Note 11.4 Graphing in Polar Coordinates

Symmetry

Symmetry Tests for Polar Graphs

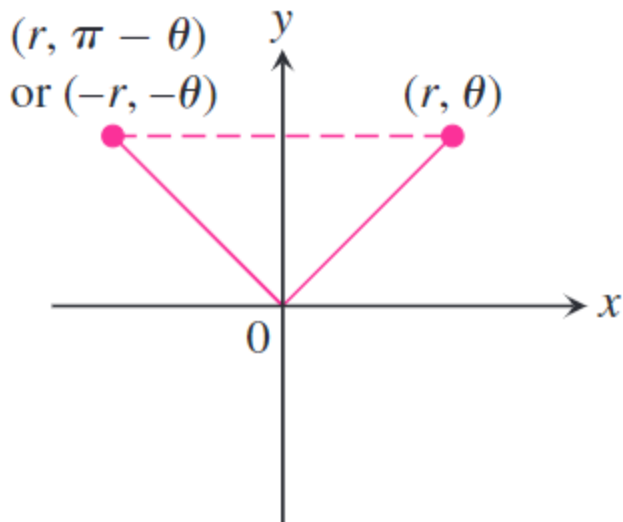
Suppose the point (r, θ)

- Symmetry about the x -axis: $(r, -\theta)$ or $(-r, \pi - \theta)$



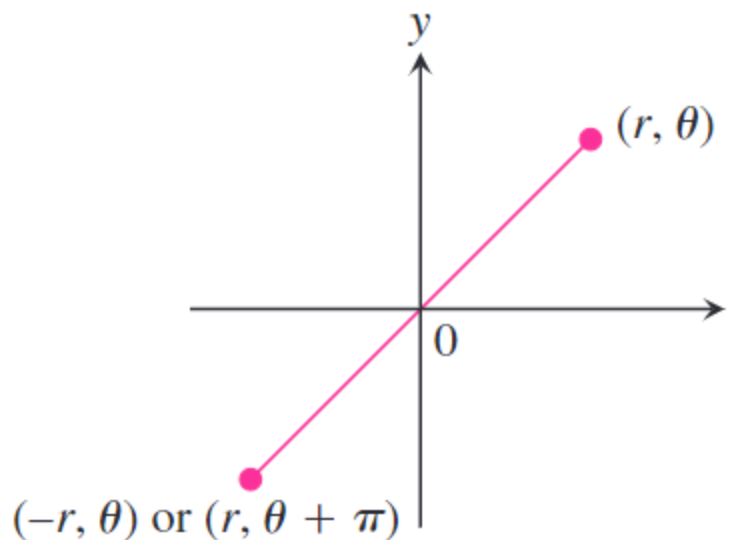
(a) About the x -axis

- Symmetry about the y -axis: $(r, \pi - \theta)$ or $(-r, -\theta)$



(b) About the y -axis

- Symmetry about the origin: $(r, \pi + \theta)$ or $(-r, \theta)$



(c) About the origin

Slope

Slope of the Curve $r = f(\theta)$

$$\left. \frac{dy}{dx} \right|_{(r,\theta)} = \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta - f(\theta) \sin \theta}$$

where $\frac{dx}{d\theta} \neq 0$ at (r, θ) .

Remarks

If the curve passes through the origin at $\theta = \theta_0$, then $f(\theta_0) = 0$ and

$$\left. \frac{dy}{dx} \right|_{(0,\theta_0)} = \frac{f'(\theta_0) \sin \theta_0}{f'(\theta_0) \cos \theta_0} = \tan \theta_0$$

Proof

The slope of a polar curve $r = f(\theta)$ in the xy -plane is still given by dy/dx rather than $r' = df/d\theta$. Then since

$$x = r \cos \theta = f(\theta) \cos \theta, y = r \sin \theta = f(\theta) \sin \theta$$

If f is differentiable function of θ and $dy/dx \neq 0$, then

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} \\
 &= \frac{\frac{d}{d\theta}(f(\theta) \cdot \sin \theta)}{\frac{d}{d\theta}(f(\theta) \cdot \cos \theta)} \\
 &= \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta - f(\theta) \sin \theta}
 \end{aligned}$$

A Technique for Graphing

- First Graphing $r = f(\theta)$ in the Cartesian $r\theta$ -plane.(default from 0 to 2π)
- Then use the Cartesian graph as a table to sketch the polar coordinate graph.